

AC Power Calculations:

$$\theta = \alpha - \beta = \angle \bar{V} - \angle \bar{I}$$

$$\bar{S} = P + jQ = S \angle \theta = (VI) \angle \theta = \bar{V} \cdot \bar{I}^*$$

$$pf = \cos \theta \quad \text{lagging if } \theta > 0, \quad \text{leading if } \theta < 0, \quad \text{unity if } \theta = 0$$

$$\bar{S}_{3\phi} = P_{3\phi} + jQ_{3\phi} = S_{3\phi} \angle \theta = (3V_{LN} I_L) \angle \theta = (\sqrt{3} V_{LL} I_L) \angle \theta = 3 \cdot \bar{S}_{LN}$$

Efficiency Calculations:

$$\eta_{TL} \text{ or } \eta_{XFMR} = (P_{OUT} / P_{IN}) \times 100,$$

$$\eta_{SYS} = (P_{LNET} / P_{SNET}) \times 100$$

Source and Load Modeling:

Ideal Source Model	Generator Model	Utility Tie Model	Load Model
- divide $\bar{Z}_G$ or $\bar{Z}_L$ by 3 if a 3ϕ generator or 3ϕ load is Δ-connected - for utility ties: $Z_s = \frac{V_{rated}^2}{S_{SC}}$ or $Z_s = \frac{V_{rated}}{I_{SC}}$ or $Z_s = \frac{V_{LLrated}^2}{S_{3\phi SC}}$ or $Z_s = \frac{V_{LLrated}}{\sqrt{3} \cdot I_{LSC}}$ and $\theta_s = \tan^{-1}(X/R)$			

Transmission Line Modeling:

Series Impedance Model	Pi Model	
		$1\phi: \bar{Z} = \bar{Z}_x + \bar{Z}_y$ $3\phi: \bar{Z} = \bar{Z}_A = \bar{Z}_B = \bar{Z}_C$
Distributed Series Impedance: $\bar{Z} = R + j\omega L$ ($\Omega / \text{length unit}$) Distributed Shunt Admittance: $\bar{Y} = G + j\omega C$ ($\mathfrak{U} / \text{length unit}$) Total Series Impedance: $\bar{Z}_{TL} = \bar{Z} \ell$ (Ω) Total Shunt Admittance: $\bar{Y}_{TL} = \frac{1}{2} \bar{Y} \ell$ ($\mathfrak{U}$) Propagation Constant: $\bar{\gamma} = \sqrt{\bar{Z} \cdot \bar{Y}}$ ($1 / \text{length unit}$) Characteristic Impedance: $\bar{Z}_c = \sqrt{\bar{Z} / \bar{Y}}$ (Ω)		

Transformer Modeling:

1 ϕ XFMR Model	3 ϕ XFMR Model	
		$a = V_{1rated} / V_{2rated}$ $\bar{Z}_x = R_1 + jX_1 = (r_1 + jx_1) + a^2(r_2 + jx_2) = Z_x \angle \theta_x$ divide $(r + jx)$ by 3 if the side is Δ $\begin{cases} \phi = 0^\circ \text{ if Y-Y or } \Delta-\Delta \text{ or } \Delta_{CBA}-\Delta_{CBA} & \phi = -60^\circ \text{ if } \Delta_{CBA}-\Delta \\ = -30^\circ \text{ if Y-}\Delta \text{ or } \Delta_{CBA}-Y & = +60^\circ \text{ if } \Delta-\Delta_{CBA} \\ = +30^\circ \text{ if Y-}\Delta_{CBA} \text{ or } \Delta-Y \end{cases}$
$\bar{Z}_x \text{ in } \Omega = \left(\frac{R_1 + jX_1 \text{ in } \%}{100} \right) \left[\frac{(\text{rated primary side KV})^2}{(\text{rated MVA})} \right]$ $\bar{Z}_x \text{ in } \Omega = \left(\frac{r_1 + jx_1 \text{ in } \%}{100} \right) \left[\frac{(\text{rated primary side KV})^2}{(\text{rated MVA})} \right] + a^2 \left(\frac{r_2 + jx_2 \text{ in } \%}{100} \right) \left[\frac{(\text{rated secondary side KV})^2}{(\text{rated MVA})} \right]$ divide $(r + jx)$ by 3 if the side is Δ		